

Race to the Beginning

Problem ID: problem-36-23668796

In the far future, scientists have built a network of time stations. Each station may contain one-way time machines which transport a traveller from one station to another.

Every time machine changes the traveller's position in time by some integer amount. A positive value moves the traveller forward in time, while a negative value moves the traveller backwards in time.

You begin at time station 0.

You may travel through the time machines as many times as you want, as long as the direction of each machine is followed.

Your goal is to determine whether it is possible to travel arbitrarily far backwards in time. In other words, for any point in the past, determine whether there is some sequence of time-machine journeys starting from station 0 which can take you even further into the past.

Input

The first line contains two integers N and M , where N is the number of time stations and M is the number of one-way time machines.

The stations are numbered from 0 to $N - 1$.

Each of the next M lines contains three integers u , v , and t .

This means there is a one-way time machine from station u to station v which changes time by t units.

A positive value of t moves the traveller forward in time, while a negative value moves the traveller backwards in time.

The constraints are:

$$1 \leq N \leq 700$$

$$0 \leq M \leq 5000$$

$$0 \leq u, v < N$$

$$-10^6 \leq t \leq 10^6$$

There may be multiple time machines between the same pair of stations.

Output

Output YES if, starting from station 0, it is possible to travel arbitrarily far backwards in time.

Otherwise, output NO.

Sample Input 1

```
4 4
0 1 3
1 2 -5
2 1 2
2 3 4
```

Sample Output 1

YES

Sample Input 2

```
3 3
0 1 4
1 2 -2
2 1 3
```

Sample Output 2

NO

Sample Input 3

```
5 3
0 1 2
3 4 -5
4 3 1
```

Sample Output 3

```
NO
```